

TECHNICAL MODEL SPECIFICATION

TasteTrace

Public Cold-Start Music and Artist Finder

Data architecture, variables, K-nearest-neighbour retrieval, diversity reranking, training design, evaluation, and production sequence.

MODEL POSITION

The current website is a deterministic prototype. This specification defines the academically explicit model and backend target without claiming that the full production pipeline has already been trained.

Prepared for Imperial College London

Big Data, AI & Machine Learning | June 2026 | Version 1.0

01 | SYSTEM DEFINITION

A public cold-start recommender

TasteTrace is designed for any visitor, including users with no account and no imported listening history. The user supplies seed tracks or artists plus optional discovery constraints. The model treats those inputs as query evidence and searches a public catalogue for content-near candidates.

Prediction task

Element	Technical definition
Input	Seed recordings or artists, natural-language brief, mood, era, geography, exploration, and diversity.
Candidate unit	A MusicBrainz recording MBID joined to metadata and feature vectors.
Primary output	A ranked and diversified set of recordings and associated artists.
Model family	Content-based KNN or ANN retrieval followed by MMR reranking.
Cold-start property	No personal listening history is required; similarity comes from supplied seeds and public data.

IMPORTANT DISTINCTION

This is not collaborative filtering. The model can later use public playlist or interaction evidence, but the core finder remains operational from content and metadata alone.

02 | FOUR-LAYER DESIGN

Source-to-recommendation architecture

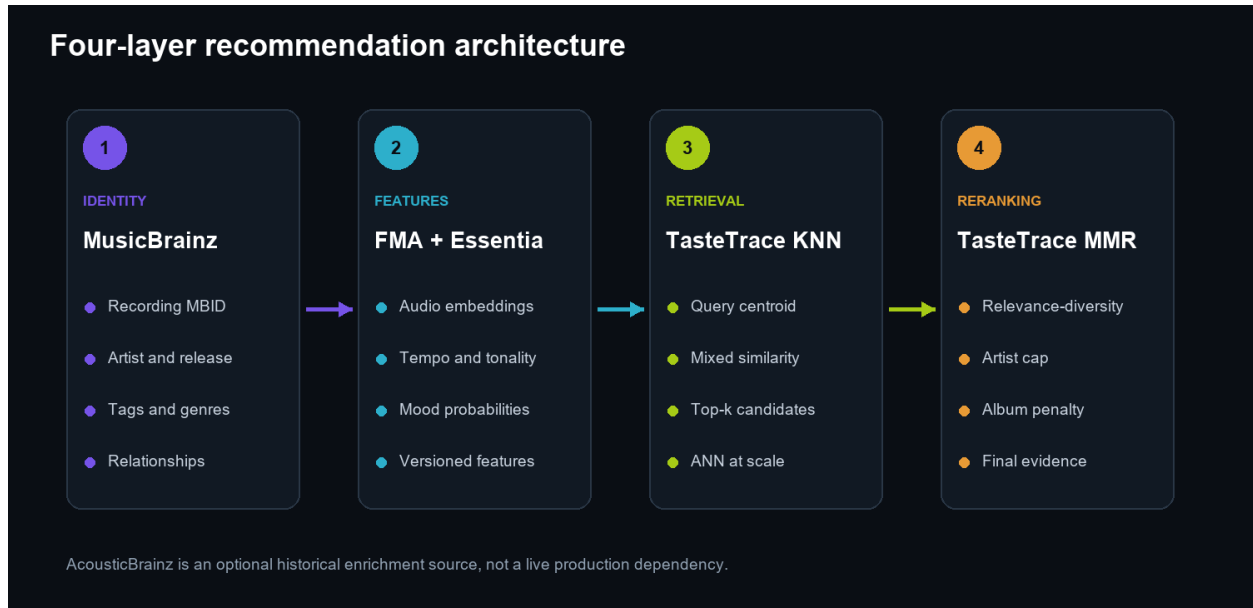


Figure 1. MusicBrainz identity, acoustic features, KNN retrieval, and MMR reranking.

MusicBrainz supplies canonical identity and catalogue relationships. FMA supplies reproducible research data. Local Essentia models create production features where lawful audio or previews are available. AcousticBrainz is treated only as an optional historical archive because its ingestion project ended in 2022.

03 | DATA RESPONSIBILITIES

What each source can and cannot claim

Source	Primary role	Do not assume
MusicBrainz	MBIDs, artist/recording/release identity, dates, countries, relationships, community tags.	Complete audio features, calibrated popularity, or preference labels.
AcousticBrainz	Archived low- and high-level descriptors linked by recording MBID.	A current or continuously expanding production API.
FMA	Research metadata, genre labels, features, play counts, and licenced audio subsets.	Full representation of commercial or global music.
Essentia / MusiCNN	Local embeddings, rhythm, tonal, spectral, genre, and mood inference.	Permission to analyse audio that the project does not lawfully access.

MusicBrainz ingestion policy

- Use recording and release-group MBIDs as stable join keys.
- Cache API responses and send an identifiable User-Agent.
- Respect public rate limits; use database dumps for large catalogue refreshes.
- Store source dates and confidence for community tags and genres.
- Keep cover-art retrieval separate through the Cover Art Archive.

04 | VARIABLES

Observation, X and optional y

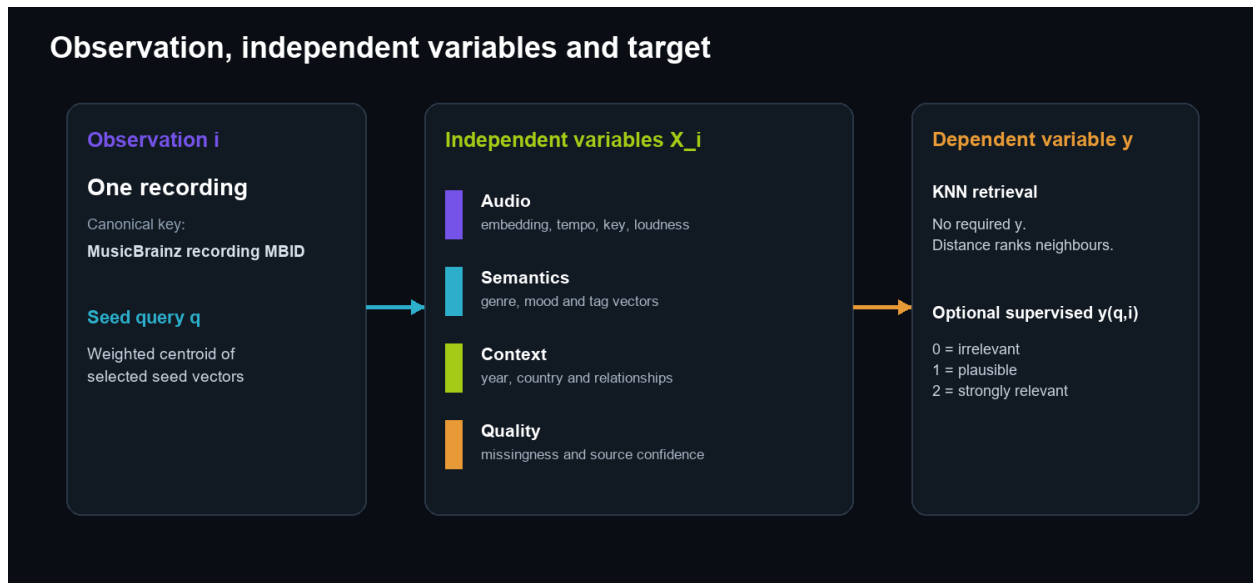


Figure 2. The recording is the observation; feature blocks form X; KNN does not require y.

```
X_i = [audio_embedding, tempo, key, loudness, mood_probs,
       tag_vector, year, country, relationships, confidence]

q = sum(w_s * X_s) / sum(w_s), for seed tracks s
```

For supervised evaluation or future learning-to-rank, $y(q,i)$ is an ordinal relevance label: 0 irrelevant, 1 plausible, and 2 strongly relevant. MusicBrainz metadata alone is not a preference target.

05 | KNN RETRIEVAL

Mixed-feature nearest neighbours

```
similarity(q,i) =  
    0.50 cosine(audio_q, audio_i)  
+ 0.20 cosine(tags_q, tags_i)  
+ 0.10 tempo_similarity  
+ 0.08 tonal_similarity  
+ 0.07 era_similarity  
+ 0.05 scene_similarity
```

The coefficients are initial product assumptions, not estimated causal effects. Validation chooses their final values. Continuous variables are standardised using training-only statistics; missing variables receive explicit flags.

Why KNN fits the product

- It directly answers the user question: what lies near these seeds?
- It supports explanations through shared features and nearest seed bridges.
- It works without user histories or account-level training data.
- It scales from exact search to approximate vector indexes.
- It can retrieve tracks or aggregate recording vectors into artist representations.

06 | ADVANCED PARAMETERS

Initial KNN and vector-index configuration

Parameter	Initial	Validation range or alternative
Candidate neighbours k	100	25, 50, 100, 200
Audio metric	Cosine	Cosine or Euclidean
Tag encoding	TF-IDF multi-hot	Binary or TF-IDF
Seed aggregation	Weighted mean	Mean, medoid, or max similarity
Metadata confidence	>= 0.50	0.30 to 0.80
Seed-artist exclusion	Enabled	Enabled or disabled
Prototype index	Exact search	Brute force
Production index	HNSW	pgvector or FAISS
HNSW M	16	8, 16, 32
HNSW ef_search	64	32, 64, 128

SCALING RULE

Exact KNN is preferred while the catalogue is small because it is easy to audit. Approximate nearest-neighbour indexing becomes appropriate at hundreds of thousands or millions of recordings.

07 | MMR RERANKING

Relevance is not the same as set quality

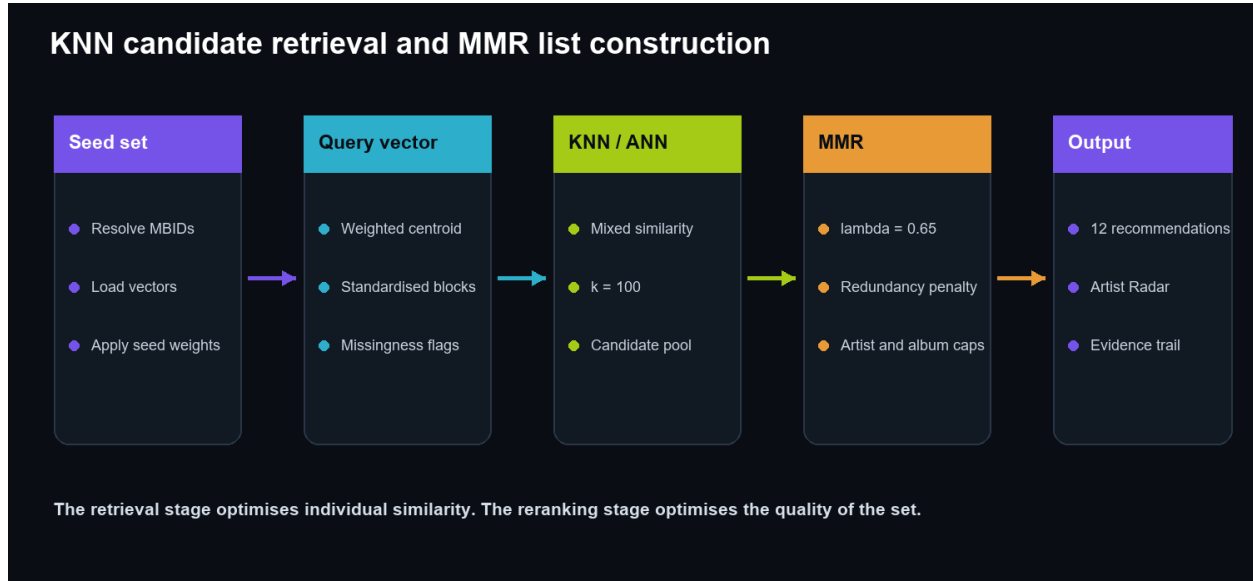


Figure 3. KNN constructs the candidate pool; MMR constructs the final recommendation set.

```

MMR(i) = lambda * relevance(q,i)
        - (1-lambda) * max_similarity(i, selected)
        - same_artist_penalty - same_album_penalty
        + emerging_artist_bonus
  
```

Parameter	Initial value	Interpretation
lambda	0.65	Balance toward relevance while retaining diversity.
Same artist penalty	0.40	Strongly suppress repeated artists.
Same album penalty	0.28	Suppress repeated releases.
Emerging coverage bonus	0.035	Small set-level exposure incentive.
Artist cap	1	Maximum selected results per artist.
Final set	12	Default user-facing recommendations.

What is trained in an instance-based model

KNN stores observations rather than fitting a conventional regression equation. The training process still fits preprocessing, selects dimensions, optionally learns a distance metric, builds the vector index, and chooses hyperparameters.

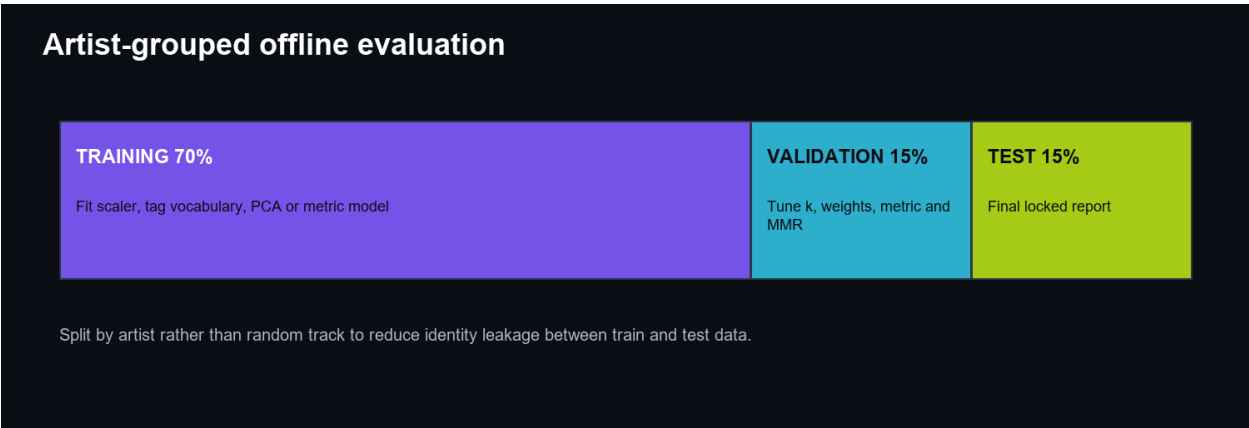


Figure 4. Proposed artist-grouped 70/15/15 split.

Dataset use

Partition	Purpose
Training	Fit scaler, vocabulary, PCA or embedding projection, and optional metric-learning model.
Validation	Choose k, feature weights, distance, HNSW parameters, MMR lambda, and penalties.
Test	Produce the final locked report once; do not tune against this set.

09 | EVALUATION

Baselines and success metrics

Required baselines

1. Popularity-only ranking.
2. Genre or tag Jaccard similarity.
3. Audio-only KNN.
4. Metadata-only KNN.
5. Hybrid KNN without MMR.
6. Hybrid KNN with MMR.

Evaluation matrix

Objective	Metrics
Relevance	Precision@K, Recall@K, MAP@K, NDCG@K.
Diversity	Intra-list diversity, unique artist ratio, unique album ratio.
Discovery	Catalogue coverage, novelty, long-tail and emerging exposure.
Reliability	Latency, cache hit rate, index build success, missing-feature rate.
Trust	Explanation usefulness and correction behaviour in a controlled study.
Representation	Exposure by country, language, genre, and catalogue segment.

NO FABRICATED ACCURACY

The project must report these as planned metrics until a labelled dataset and locked test evaluation have actually been run.

Production services and batch sequence

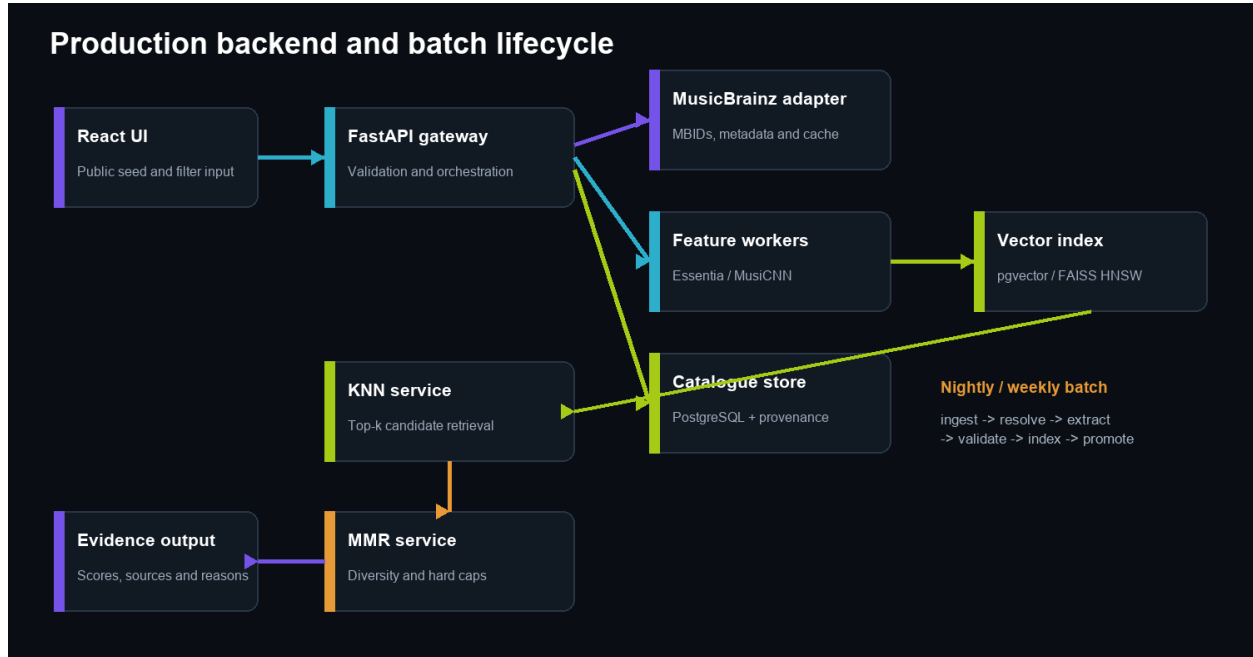


Figure 5. Online recommendation services plus a versioned offline feature and index pipeline.

Controlled batch release

1. Ingest MusicBrainz metadata or database-dump changes.
2. Resolve recording and release identities through MBIDs.
3. Attach FMA, archived AcousticBrainz, or local Essentia features.
4. Validate schema, missingness, licence, and provenance.
5. Fit versioned preprocessing and rebuild the vector index.
6. Run relevance, diversity, representation, and latency checks.
7. Promote the index only when acceptance thresholds pass; retain the previous version for rollback.

11 | IMPLEMENTATION BOUNDARY

Current prototype versus target system

Capability	Current application	Specified target
Catalogue	83 curated tracks	MusicBrainz-backed catalogue and cache
Audio variables	Deterministic tag-derived values	Essentia embeddings and descriptors
Retrieval	Full-catalogue weighted scoring	KNN or ANN candidate retrieval
Training data	None	FMA plus labelled similarity tasks
Dependent variable	None	Optional ordinal relevance label
Diversity	MMR-style deterministic reranking	Validated MMR and exposure monitoring
Backend	Browser-only	FastAPI, PostgreSQL, vector index, workers
Updates	Code release	Versioned batch and rollback pipeline

Technical risks

- MusicBrainz tags and genres are community-generated and incomplete.
- FMA is a research corpus, not a complete market representation.
- AcousticBrainz coverage is historical and frozen.
- Audio extraction requires lawful access to audio or previews.
- Similarity should never be presented as objective artistic quality.
- All model and index versions require source, date, licence, and transformation provenance.

APPENDIX | REFERENCES

Primary technical sources

Reference	Project use
MusicBrainz API	Canonical identity, metadata search, relationships, tags, and genres.
MusicBrainz database downloads	Bulk catalogue refresh at scale.
AcousticBrainz	Optional historical descriptors; discontinued ingestion noted.
FMA dataset	Reproducible feature, genre, training, and evaluation corpus.
Essentia models	Local embeddings, classification, auto-tagging, and descriptors.
Carbonell and Goldstein (1998)	Maximal marginal relevance diversification.

URLs

- https://musicbrainz.org/doc/MusicBrainz_API
- https://musicbrainz.org/doc/MusicBrainz_Database/Download
- <https://acousticbrainz.org/>
- <https://github.com/mdeff/fma>
- <https://essentia.upf.edu/models.html>

REPOSITORY DOCUMENT

The full text specification is also available in docs/MODEL_TECHNICAL_SPECIFICATION.md.